Design Decision Matrix

Fillable PDF. Click a blue box to type; tick the small squares. Save with **File › Save** and attach the saved PDF to your assignment. Works best in the free Adobe Acrobat Reader; macOS Preview and web browsers also work. If an answer does not fit its box, continue it on an extra page. Source: `Step_04_Study_Architecture_and_Experimental_Design/design_decision_matrix.md` (rebuild with `tools/make_fillable_pdfs.py`).

Use this matrix to select the appropriate study design given your construct type, research goal, and practical constraints. Circle or highlight the row that matches your situation and carry that design code into `study_design_checklist.md`.

Key Terms

• **BACI (Before-After Control-Impact):** A classic experimental design where you measure both a treatment group (Impact) and a comparison group (Control) both *before* and *after* an intervention. This allows you to account for pre-existing differences and environmental changes over time.

• **ACI (After Control-Impact):** A design where measurements are only taken *after* the intervention across both a treatment and control group. Used when baseline (before) measurements are impossible or would tip off participants.

Part 1: Construct × Design Compatibility

Construct stability	Research goal	Intervention present?	Recommended design	Notes
Trait-like (values, worldviews)	Describe population distribution	No	**Cross-sectional**	Single snapshot is sufficient
Trait-like (values, worldviews)	Detect change over time	No	**Longitudinal panel (multi-year)**	Weeks are insufficient; plan for years
Attitude (policy support, wildlife attitudes)	Describe and predict	No	**Cross-sectional**	Add covariates for temporal context
Attitude	Test causal effect of stimulus	Yes — pre-measurement feasible	**BACI**	Gold standard; randomize if possible
Attitude	Test causal effect — pre-measurement impossible or sensitizing	Yes	**ACI / natural experiment**	Cannot estimate within-individual change
State (affect, mood, situational emotion)	Capture real-time fluctuation	No	**Experience sampling / diary**	Single survey misses variation
State	Test immediate effect of stimulus	Yes	**Within-subjects BACI**	Control order effects; short inter-trial interval
Mixed (attitude + affect)	Test stimulus effect	Yes	**BACI with immediate + delayed post**	Separate timing for state vs. attitude outcomes

---

Part 2: Intervention Design Decision Tree

```
Is there a stimulus / treatment?
│
├─ NO → Cross-sectional or longitudinal panel
│         └─ Is the construct stable (trait-like)?
│               ├─ YES → Single wave cross-sectional is valid
│               └─ NO  → Repeated measures panel required
│
└─ YES
      │
      ├─ Is pre-measurement feasible?
      │     │
```

```
│     ├─ YES → BACI
│     │         ├─ Can participants be randomly assigned? → RCT-BACI (strongest)
│     │         └─ No random assignment possible?       → Quasi-experimental BACI
│     │
│     └─ NO (pre-measurement impossible OR would sensitize)
│               └─ ACI / Natural experiment
│                   └─ Argue baseline equivalence via matching or auxiliary data
│
└─ Is the expected change DURABLE or TRANSIENT?
       ├─ Durable  → Include delayed follow-up (T2 at ≥ 3 months)
       └─ Transient → Immediate post only; note decay expected
```

Part 3: Within-Subjects vs. Between-Subjects Trade-offs

*Note on **Washout**: A washout period is a designated gap in time between treatments in a within-subjects design. It allows the psychological or behavioral effects of the earlier treatment to dissipate so they do not "carry over" and contaminate the measurement of the next treatment.*

Criterion	Within-subjects	Between-subjects
Statistical power	Higher (participant = own control)	Lower (requires larger N)
Testing / carry-over effects	**Vulnerable** — must counterbalance or use washout	Not applicable
Attrition risk	Higher (multiple contacts required)	Lower
Demand characteristics	Higher (purpose more transparent)	Lower
Suitable for	State constructs, short studies	Trait/attitude constructs, sensitive topics
Recommended mitigation	Counterbalancing; filler tasks; long washout	Random assignment; covariate balance check

---

Part 4: Temporal Context Risk Assessment

How to use this section: Human attitudes and behaviors do not exist in a vacuum; they interact closely with transient environmental and social events. If external events—like a highly publicized local human-wildlife conflict or unseasonable weather—coincide with your data collection, they can severely confound the signals you are trying to measure. Use the table below BEFORE launch to actively map out these risks. Rate each one based on its likelihood to overlap with your study. If you mark any factor as Medium or High, commit to the recommended action to either statistically control for the event (via covariates) or chronologically avoid it.

Rate each factor Low / Medium / High for your data collection window.

Factor	Risk level	Action if Medium/High
Data collection during hunting / culling season		Record as covariate; consider delaying or stratifying by season
Active depredation conflict or rewilding event nearby		Document events; add event-proximity covariate
National or regional media spike on focal species		Record Google Trends or media index; add as covariate
Extreme weather affecting outdoor activity		Record daily weather; assess mood confounds
Major policy announcement (listing, delisting, quota change)		Flag wave; consider ACI framing

---

Part 4b: Discrete Choice Experiments (DCE) — a stated-preference alternative to BACI

When to use: DCEs ask respondents to repeatedly choose between hypothetical alternatives (e.g., management scenarios, land-use options, policy packages) that vary systematically across attributes and levels. Unlike a single-item willingness-to-pay question, a DCE decomposes preferences into the contribution of each attribute (e.g., cost, distance, species affected), and is well suited to socio-ecological trade-off questions (habitat restoration design, protected-area management, payment-for-ecosystem-service scheme design) where no real-world BACI intervention is feasible.

Consideration	Guidance
Attribute and level selection	Ground in qualitative/expert pre-work (focus groups, Step 1) before fielding; over-large attribute sets increase cognitive burden and attribute non-attendance
Attribute non-attendance (ANA)	Respondents may ignore one or more attributes when making choices, biasing willingness-to-pay estimates if unaddressed; test for and model ANA explicitly (Notaro et al., 2022)
Experimental design	Use an efficient (D-optimal or Bayesian-efficient) design rather than a full factorial to keep the number of choice tasks tractable
Model choice	Conditional/mixed logit is standard; report willingness-to-pay space vs. preference space estimates and justify the choice
Review of applications	See Cop & Njavro (2022) for a systematic review of DCE applications specifically in agricultural/environmental risk management, useful for benchmarking attribute choices against the published literature

---

Part 5: Selected Design Summary (fill in)

Field	Your entry
Focal constructs	
Stability classification	☐ Trait ☐ Attitude ☐ State
Intervention present	☐ Yes ☐ No
Pre-measurement feasible	☐ Yes ☐ No ☐ Partially
Design type selected	☐ Cross-sectional ☐ BACI ☐ ACI ☐ Within-subjects ☐ Between-subjects ☐ Mixed
Expected change type	☐ Durable ☐ Transient ☐ N/A
Post-measurement occasions	T1: T2:
Temporal context risks	☐ Low ☐ Medium ☐ High
Pre-registration platform	☐ OSF ☐ AsPredicted ☐ Other:

Cross-reference: `study_design_checklist.md` · `design_docs/randomization_plan.md` · `../Step_05_PreRegistration/` · `../Step_09_Power_and_Sampling/`

References: Notaro, De Salvo, & Raffaelli (2022); Cop & Njavro (2022).